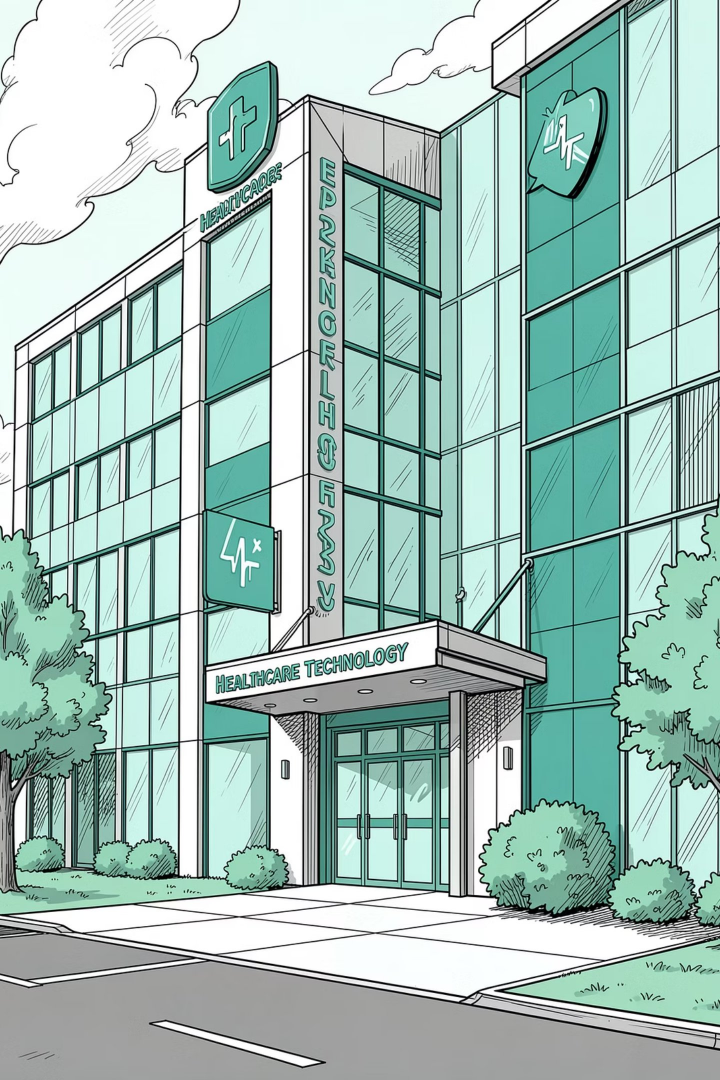




Data Engineering Internship — CitiusTech

Project Team: Arcadia Data Management • Duration: 6 months • Presented
by: Tejas Desai



About CitiusTech



Who they serve

Healthcare providers, payers, medtech and life sciences — specialised in clinical and operational data solutions.



Core capabilities

Healthcare data platforms, analytics, cloud transformation and AI-driven solutions for enterprise scale.



Commitment

Security, compliance and trusted delivery across regulated healthcare environments.



Internship Objective & Learning Goals

Gain hands-on experience in healthcare data engineering: build cloud-native data pipelines, apply validation & ETL best practices, and contribute to POCs that solve real business problems.

- Practical cloud data processing (AWS)
- End-to-end pipeline development (ETL/ELT)
- Exposure to AI/LLM integrations for operational knowledge

Onboarding & Training

Domain & Tools

Healthcare domain induction,
Python for data engineering,
PySpark fundamentals.

AWS & Data Platforms

S3 data lake patterns, Athena,
Redshift, Serverless
orchestration, Kinesis basics.

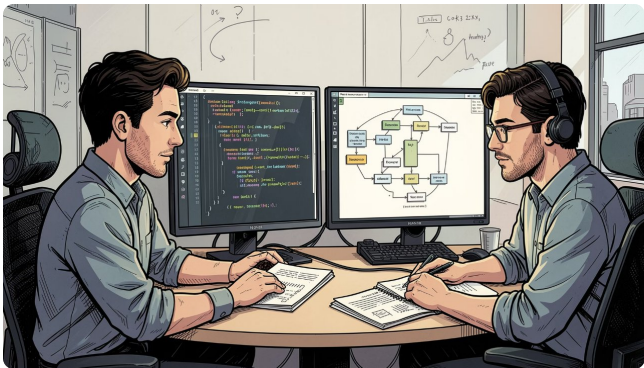
Modern Stack

DBT, Snowflake concepts, SQL best practices, CI/CD for data engineering.

Daily hands-on sessions, assignments, and continuous mentor feedback accelerated learning.



Team Structure & Work Culture

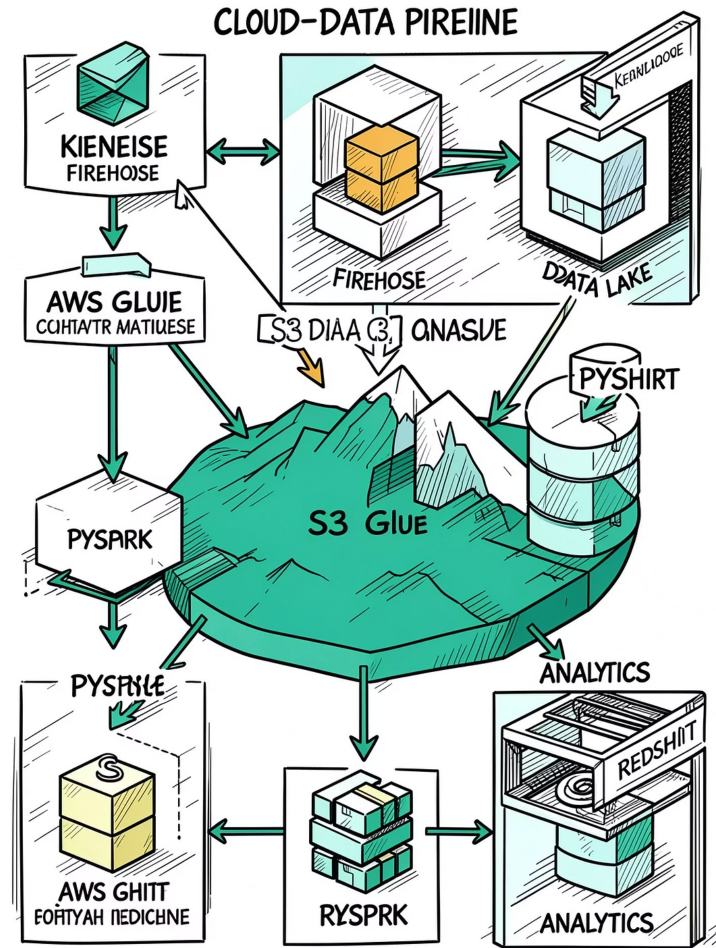


Arcadia Data Management: cross-functional team with assigned managers and mentors. Cadence included daily stand-ups, task tracking and design reviews.

- Agile methodology and sprint-based delivery
- Close collaboration with Data Engineers & Architects
- Emphasis on clear requirements and professional communication

Healthcare Data Engineering Architecture

Orchestration: AWS Step Functions. Technologies used: S3, Kinesis, Glue (PySpark), Redshift, Athena, Step Functions, SQL. This architecture supports high-throughput ingestion, scalable ETL, and analytic-ready tables.





Work In Progress — Validation Module

Contributed to the WIP stage of the pipeline focusing on customer data validation. Responsibilities included:

- Designing Python-based validation scripts and rules
- Implementing data quality checks (missing fields, schema validation, range checks)
- Flagging and classifying incorrect or incomplete records for downstream handling



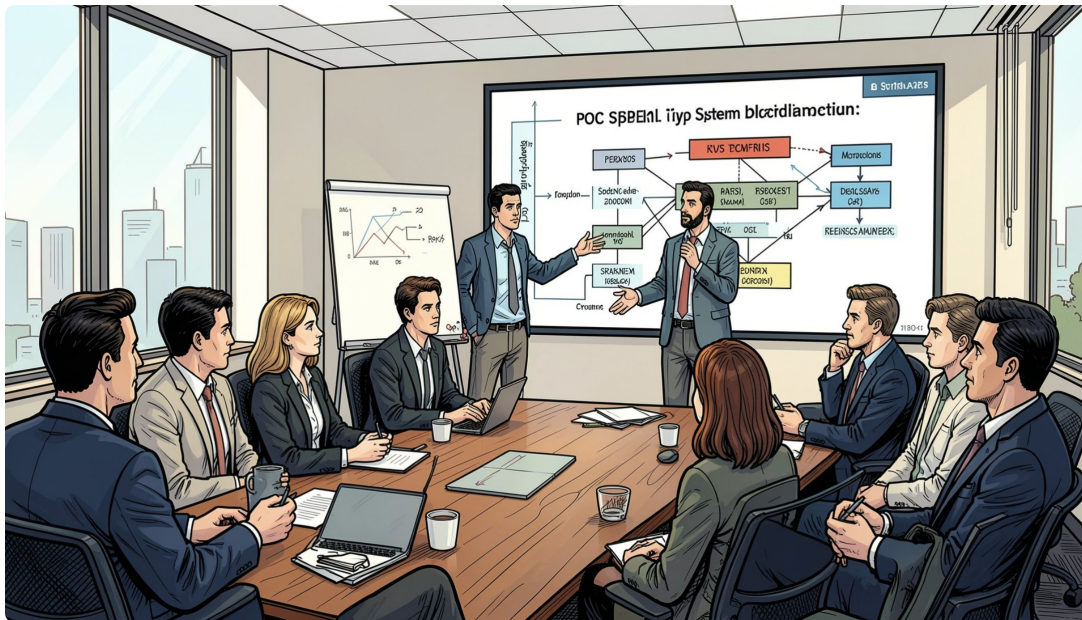
Impact: Increased reliability of downstream processes and reduced data-quality incidents entering analytics.

Proof of Concept — AI-powered RAG Solution

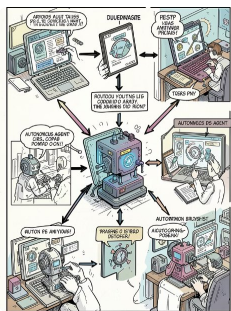
Objective: Build a Retrieval-Augmented Generation system to surface operational knowledge quickly from logs and documentation.

- Ingested sources: AWS logs, Confluence pages
- Pipeline: chunking → embeddings → FAISS vector store → retriever → LLM
- Features: semantic search, context-aware retrieval, question-answering interface

Technologies: Python, LangChain, FAISS, LLM APIs. Outcome: Demonstrated feasibility and received leadership feedback for next-phase integration.

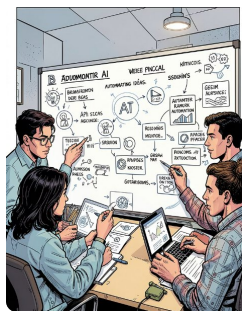


AI Agent Research & Contributions



Use-case exploration

Mapped agentic automation opportunities for routine operational tasks and runbook execution.



Knowledge transfer

Prepared concise research notes and walkthroughs to help the team understand Agent workflows and constraints.

Outcome: Improved team readiness to evaluate agentic automation in future projects.

Skills Gained & Key Takeaways

Technical

Python, PySpark, SQL, AWS (S3, Glue, Redshift, Athena), Snowflake, DBT, ETL/validation, RAG systems.

Professional

Agile delivery, cross-team collaboration, requirement analysis, presentation & stakeholder communication.

Summary: Developed strong fundamentals in cloud-based healthcare data engineering, improved pipeline reliability through validation efforts, delivered a successful RAG POC, and contributed to AI adoption discussions.

- Next steps: Expand POC into staging, integrate retriever with runbooks, and formalise validation rule library.

Thank you — Questions & Answers

